

P r e d i c t i n g

S y r i a T e l

Using Machine Learning for Strategic Retention

P R E S E N T A T I O N



B u s i n e s s

K e y

S y r i a T e l
t e a m s

S t r a t e g i c

S y r i a T e l
l e a d i n g v e n u e a n d
c o s t s a c q u i r i n g
c u s t o m e r s

The goal is to develop a predictive
model that identifies at-risk customers
early.



T a r g e t e d



I m p r o v e d



P e r s o n a l i z e d

A m i n i m i z i n g ,
c u s t o m e r s l o s t .

D a t a

D a t a

S y r i a I T e l

Historical Data

3

D a t a s e t

Records with zero missing values



F e a t u r e s

Behavioural and service-related
variables



P r e d i c t i v e

Indicators ready for churn modeling

Data

1

Feature

Removed irrelevant features such as phone number and state to focus on predictive signals.

2

Encoding

Encoded categorical variables like international plan and voicemail plan; split data into training and testing sets.

3

Scaling

Applied feature scaling for Logistic Regression and prevented data leakage by fitting transformations only on training data.

M o d e l i n g

PHASE 1

B a s e l i n e

L o g i s t i c

- Simple and interpretable
- Provided a performance benchmark
- High accuracy but poor recall for churners

PHASE 2

Improved

Decision Tree

- Captures non-linear relationships
- Significantly improved recall
- Better balance between precision and recall

PHASE 3

R e f i n e d

T u n e d

- Hyperparameters adjusted to control complexity
- Aimed to reduce overfitting
- No significant recall gain over base model

Logistic

Benchmark

This benchmark
why

It
achievement, poorly
identifying

BASELINE RECALL

0

△ Poor

IMPROVED RECALL

0

significant

Decision

Capturing

This churn
objective

It Decision's to
capture ~~linear~~ and
better precision.

T u n e d

C o m p l e x i t y

R e f i n e m e n t m a n a g e
c o m p l e x i t y u c e .

It
improves beyond



M o d e l

Overfitting Protection Achieved



0 A c c u r a c y

E v a l u a t i o n

Accuracy



Precision



R E C A L L



P R I M A R Y

F1-score



W h y

M i s s i n g
r e v e n u e

l o s i n g

w i t h o u t

d i r e c t l y

M o d e l

P e r f o r m a n c e

M o d e l	A c c u r a c y	P r e c i s i o n	R e c a l l	F 1
L o g i s t i c	0	0	0	0
D e c i s i o n	0	0	0	0
T u n e d	0	0	0	0

C o n c l u s i o n

The decision provides the best
balance between
the most for
problem .

B u s i n e s s



R i s k

Use the Decision Tree model to identify customers at high risk of churn before they leave the network.



R e t e n t i o n

- Personalized offers
- Proactive customer support
- Service or pricing adjustments



R e v e n u e

Focus on early identification to reduce customer attrition and protect corporate revenue.

L i m i t a t i o n s

01

C o n s t r a i n t s

Based on historical data; may not capture future behavior changes.

02

E x t e r n a l

External factors like competitor pricing are not currently included.

03

C l a s s

Slight class imbalance may affect specific model performance segments.

N e x t



A d v a n c e d

Explore Random Forest and Gradient Boosting to further refine prediction accuracy.



D a t a

Perform feature engineering and incorporate additional data sources for a 360-degree view.



O p e r a t i o n a l i z a t i

Deploy model for real-time churn monitoring and continuously retrain as behavior evolves.

A blue-tinted photograph of two women in business attire shaking hands in an office setting. In the background, there are computer monitors displaying charts and graphs, and large windows looking out onto a city skyline.

A N Y

T H A N K

